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OF QUEENSLAND
A U S T R A L I A

NSW ELECTORAL EXPENDITURE ANALYTICS

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1. Introduction

This project analyses the NSW Electoral Commission disclosure data to examine how political parties allocate their electoral expenditure across categories and over time, with the objective of improving transparency and interpretability for stakeholders.

Electoral spending attracts significant attention from stakeholders, from voters to media outlets. However, the allocation of electoral expenditure by parties raises public scrutiny and concerns regarding fairness. For example, the two major parties, the Liberal Party and the Labor Party, collectively spent around AUD 353 million during the last federal election cycle, highlighting the scale of money involved [1]. The findings are expected to provide better transparency to voters and offer insights that can support electoral spending policies.

For this project, the focus will be on NSW electoral expenditure, using data from the NSW Electoral Commission's disclosures, to maintain a manageable scope. Although electoral expenditure data is accessible, it is often complex and difficult to interpret, making it challenging for stakeholders to understand how money is allocated.

The project transforms and summarises the data and presents the findings using dashboards to provide an overview of spending patterns and address these issues. This report initially examines existing approaches to electoral expenditure analysis, identifies key limitations, reviews related work, outlines the objectives and proposed methodology, and presents preliminary exploratory data analysis to demonstrate initial findings.

2. Current Situation

Current approaches to analysing electoral expenditure rely on publicly available disclosure data submitted by political parties. 'Electoral expenditure' refers to expenditure for or in connection with promoting or opposing a political party, the election of a candidate or candidates, or influencing the voting at an election [2] and is monitored to ensure transparency and fairness in the democratic process.

In practice, these disclosures are often presented as spreadsheets containing thousands of raw, transaction-level records. As a result, common analytical approaches focus on summarising this data using averages and visualisations, including line charts and histograms, to identify general trends across parties and time. These approaches are widely used because they reduce data complexity and provide a high-level overview of spending.

However, these approaches still have limitations, as they do not clearly show how expenditure is distributed across categories such as advertising, research, staffing, and operational costs, limiting stakeholders' ability to draw insights and make informed decisions.

These limitations highlight the need for detailed examination of challenges in current approaches to electoral expenditure analysis, which will be discussed in the following section.

3. Problem Identification

Current approaches to analysing electoral expenditure have several limitations that reduce their effectiveness for stakeholders. These approaches rely heavily on charts and graphs, which may not be intuitive to interpret and can sometimes be misleading when presented without sufficient context. Previous studies indicate that graphical representations may generate a misleading impression of objectivity, potentially concealing important nuances within the data [3].

In addition, existing approaches tend to focus on overall expenditure rather than examining how funds are allocated across specific categories such as advertising, research, and administrative costs. The disclosure data is also complex, consisting of thousands of transaction-level records, which further limits interpretability.

As a result, stakeholders are unable to identify spending patterns or compare allocation strategies across parties. This lack of interpretability reduces the effectiveness of electoral expenditure data in supporting transparency, accountability, and informed decision-making.

To better understand how these limitations have been addressed in previous studies, the following section reviews existing studies on electoral expenditure analysis.

4. Related Work

Previous studies have examined the relationship between campaign spending and electoral outcomes using quantitative methods. For example, [5] applies data envelopment analysis (DEA) to evaluate the efficiency of campaign spending in generating vote share. The study finds that higher efficiency in campaign spending leads to better electoral performance and that both candidate and district characteristics influence this relationship.

More recent studies have explored the use of data analysis and visualisation techniques to improve the accessibility and interpretability of electoral data. For example, [6] demonstrates how combining

data processing and visualisation methods can reveal meaningful patterns while highlighting challenges related to data availability, and accessibility.

However, these approaches primarily focus on measuring efficiency or analysing electoral outcomes, rather than examining how expenditure is distributed across specific categories such as advertising, and staffing. Additionally, many of these studies rely on complex modelling techniques, which limits interpretability for stakeholders.

This project builds on existing work by focusing on category-level expenditure analysis and developing interpretable visualisations and dashboards, enabling stakeholders to better understand electoral spending patterns.

Building on these identified gaps and insights, the following sections outline the objectives of this project and the proposed approach to address these limitations.

5. Objectives

The project aims to achieve the following objectives:

- Aggregate electoral expenditure data into categories, such as advertising, research, staffing, and operational costs.
- Enable comparison of spending patterns across parties.
- Analyse temporal trends in electoral expenditure to identify changes in campaign strategies.
- Detect anomalies in expenditure, particularly those arising from high-value transactions.
- Develop an interactive dashboard that presents electoral expenditure data in a clear and interpretable way.
- Develop a predictive model to forecast electoral expenditure trends across parties and categories using historical data.

To achieve these objectives, a structured approach is required to transform raw electoral expenditure data into meaningful and interpretable insights.

6. Proposed Solution

The project proposes a structured approach to transform electoral expenditure data into a more interpretable and analysable format. The approach involves collecting disclosure data across multiple

years and preprocessing it by removing irrelevant information, handling missing values, and merging datasets, as disclosures are made annually (1 July to 30 June) [4].

The processed data is aggregated at both the party level and the category level to examine overall expenditure trends and how spending is distributed across categories such as advertising, research, staffing, and operations. The analysis identifies patterns in how expenditure is allocated over time and compares spending strategies across parties.

The results are presented using an interactive dashboard developed in Power BI, which allows users to explore data using filters and category selections, enabling deeper analysis of spending patterns across parties and time periods. This interactive approach enhances usability and allows stakeholders to generate customised insights based on their interests, making electoral expenditure data more accessible.

In addition to descriptive analysis, the project incorporates a predictive modelling component to forecast electoral expenditure trends over time. Using a time-based train-test split, historical expenditure data is used to model and predict future spending patterns across parties and categories. The predictive component complements the exploratory analysis by extending insights beyond historical patterns and supporting more informed decision-making.

Overall, this approach offers advantages over existing methods by focusing on clarity and meaningful insights, making the data more interpretable to a wider range of stakeholders, including voters and policymakers.

To implement this approach, the following sections describe how the project extracts, processes, analyses, and models the data, and outline the project timeline.

7. Methodology

This project adopts a data-driven approach to analyse electoral expenditure patterns using disclosure data from the NSW Electoral Commission.

7.1 Data Extraction and Integration

Electoral expenditure data is extracted from PDF disclosures using parsing techniques implemented with the pdfplumber library. For each document, relevant tables are identified and extracted, and disclosure sections (Part A-H) are mapped to corresponding categories such as advertising, research, staffing, and operations.

The extracted datasets for each political party are then combined into a single unified dataset to enable consistent analysis across parties and time periods.

7.2 Data Cleaning and Preprocessing

The raw data is pre-processed to ensure consistency and accuracy. This includes removing non-numeric characters from expenditure values, converting them into numerical format, standardising date formats, and handling missing or invalid entries. Records with missing key variables such as amount or date are removed to maintain data quality.

7.3 Feature Engineering and Aggregation

Key features are derived from the dataset, including expenditure categories and monthly periods. The data is aggregated at both the party level and category level to analyse total expenditure and the distribution of spending across categories. Additional aggregations are performed to identify high-value transactions that may influence summary statistics.

7.4 Exploratory Data Analysis

Exploratory data analysis is conducted to identify patterns in electoral expenditure across parties, categories, and time. This includes analysing total spending, category-level distributions, transaction-level distribution, and temporal trends. Distributional analysis is used to identify skewness and the impact of large transactions on interpretability.

7.5 Visualisation and Dashboard Development

The analysed data is visualised using charts and graphs to highlight key patterns. An interactive dashboard is developed in Power BI, enabling users to explore expenditure data using filters and category selections. This enhances interpretability and allows users to generate customised insights across parties and time periods.

7.6 Predictive Analysis

To extend beyond descriptive analysis, the project incorporates a predictive modelling component to forecast electoral expenditure trends. A time-based train-test split is used, where historical data is used to model and predict future expenditure patterns across parties and categories. This component provides a forward-looking perspective and complements the exploratory analysis.

8. Project Timeline

This project is conducted in multiple stages, progressing from data collection to analysis and final reporting. The timeline is structured as follows:

- **Week 1-2:**
Data collection and extraction from NSW Electoral Commission disclosure PDFs, followed by initial data cleaning and preprocessing.
- **Week 3:**
Data integration and feature engineering, including categorisation of expenditure and creation of time-based variables.
- **Week 4-5:**
Exploratory data analysis to identify patterns in expenditure across parties, categories, and time, along with development of initial visuals.
- **Week 6:**
Development of interactive dashboard in Power BI to present key insights and improve interpretability.
- **Week 7:**
Implementation of predictive analysis using time-based modelling to forecast expenditure trends.
- **Week 8:**
Refinement of analysis, including handling of outliers and validation of findings.
- **Week 9:**
Finalisation of dashboard, integration of insights, and preparation of presentation.
- **Week 10:**
Final report writing, documentation, and project submission.

9. Exploratory Data Analysis

Preliminary exploratory data analysis (EDA) was conducted to identify key patterns in electoral expenditure across parties, categories, and time. The analysis focuses on three major political parties (Liberal, Labor, and National) and is based on electoral expenditure data from July 2022 to March 2023.

In this section, the term campaign expenditure is used to refer to total reported spending, including expenditures incurred by both political parties and associated candidates.

9.1 Total Expenditure by Party

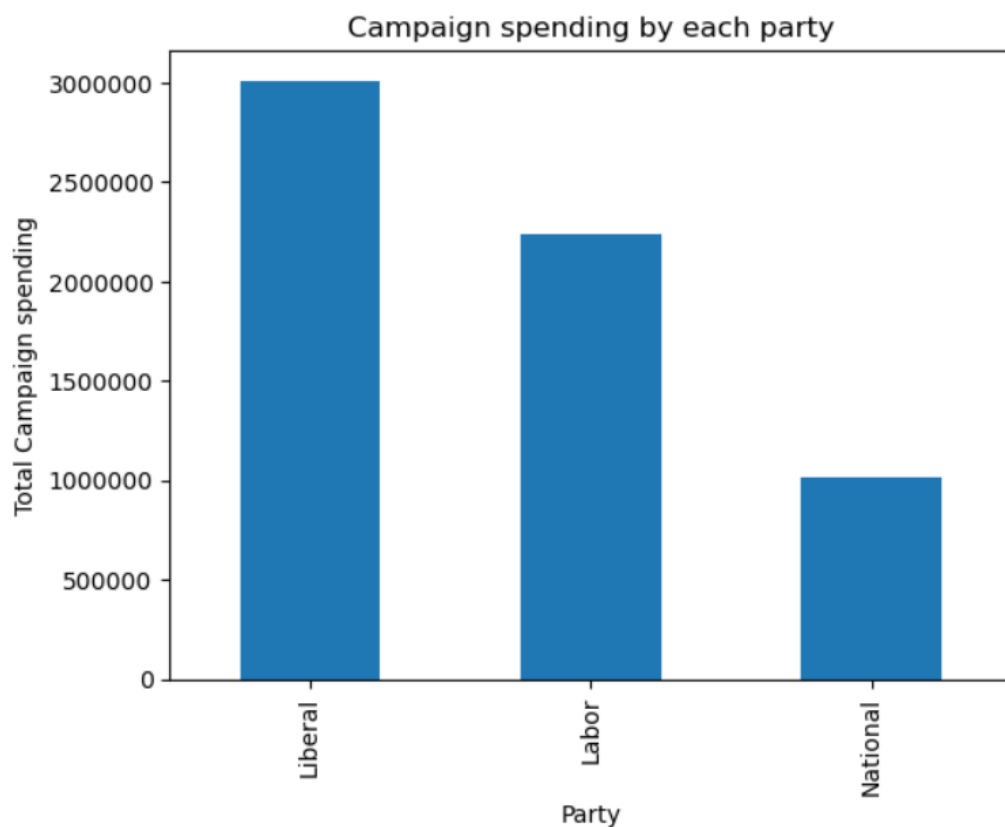


Figure 1: Total campaign expenditure by party

The Liberal Party recorded the highest total expenditure (approximately AUD 3.01M), followed by Labor (AUD 2.24M) and National (AUD 1.01M). This indicates differences in campaign expenditure scale across parties. However, this result is influenced by a small number of high-value transactions, suggesting that total expenditure may not reflect typical spending behaviour.

9.2 Distribution of transaction values

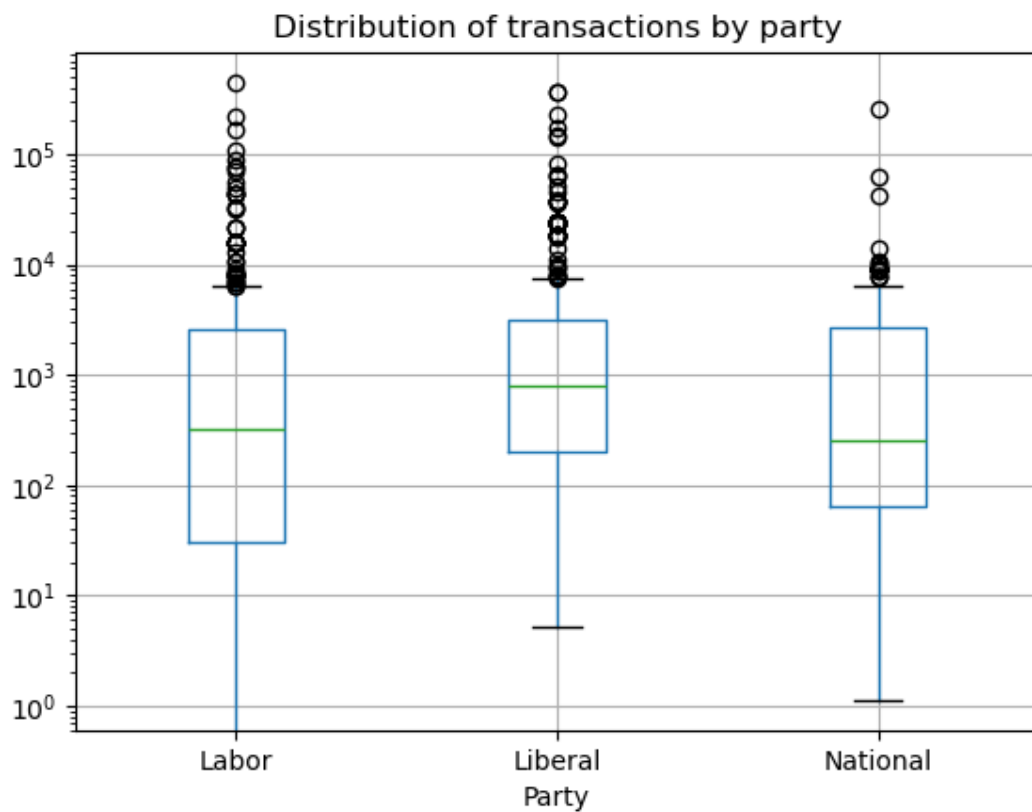


Figure 2 : Distribution of Transaction Values by Party

The boxplot shows that expenditure data is highly skewed for all parties. A logarithmic scale is used to better visualise the distribution due to the presence of large outliers. Most transactions are relatively small, while a few extremely large transactions act as outliers. This skewness indicates that mean values may not be reliable and highlights the need for outlier-aware analysis.

9.3 Spending Distribution Across Categories

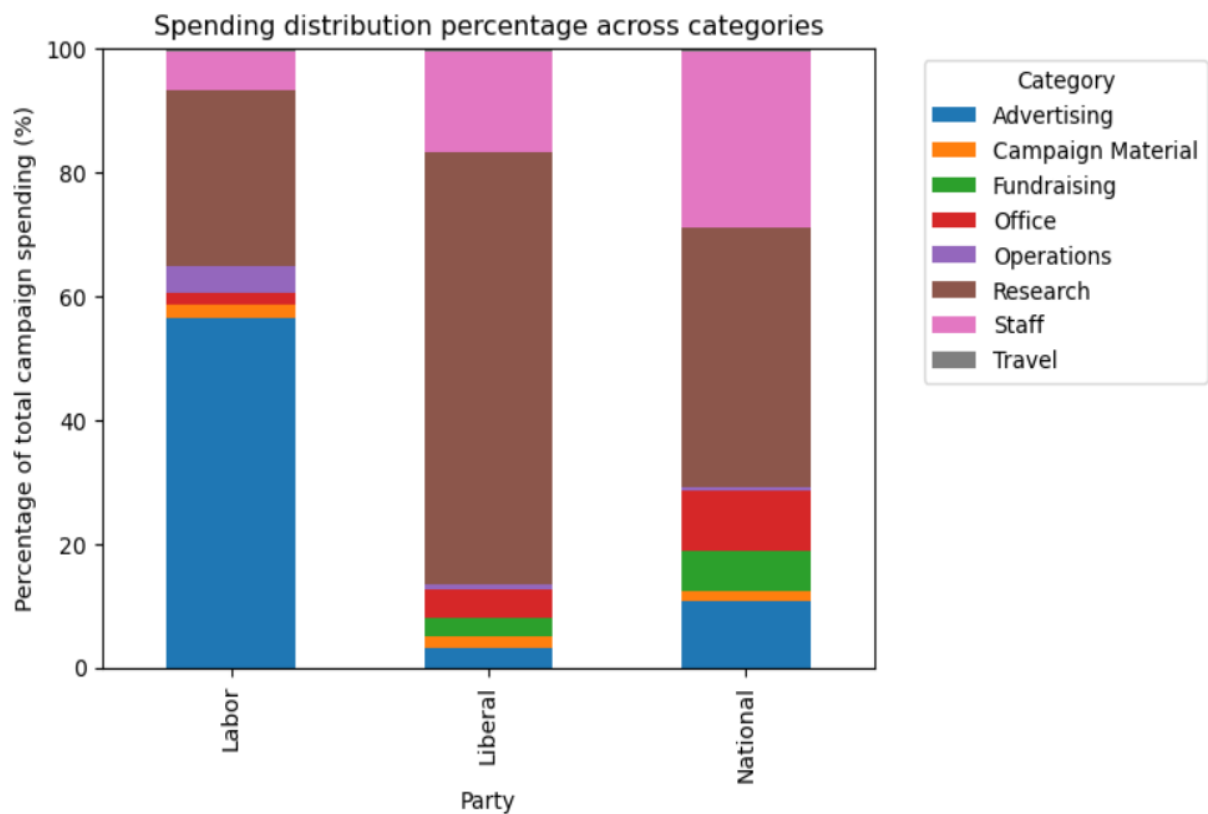


Figure 3 : Percentage distribution of spending across categories

Spending patterns vary significantly across parties. Labor allocates approximately 56% of its expenditure to advertising, while the Liberal Party allocates 70% to research. The National Party shows relatively higher proportions in staffing and operational costs.

However, these distributions may be influenced by large individual transactions, which can distort category-level comparisons.

9.4 Time-Based Spending patterns

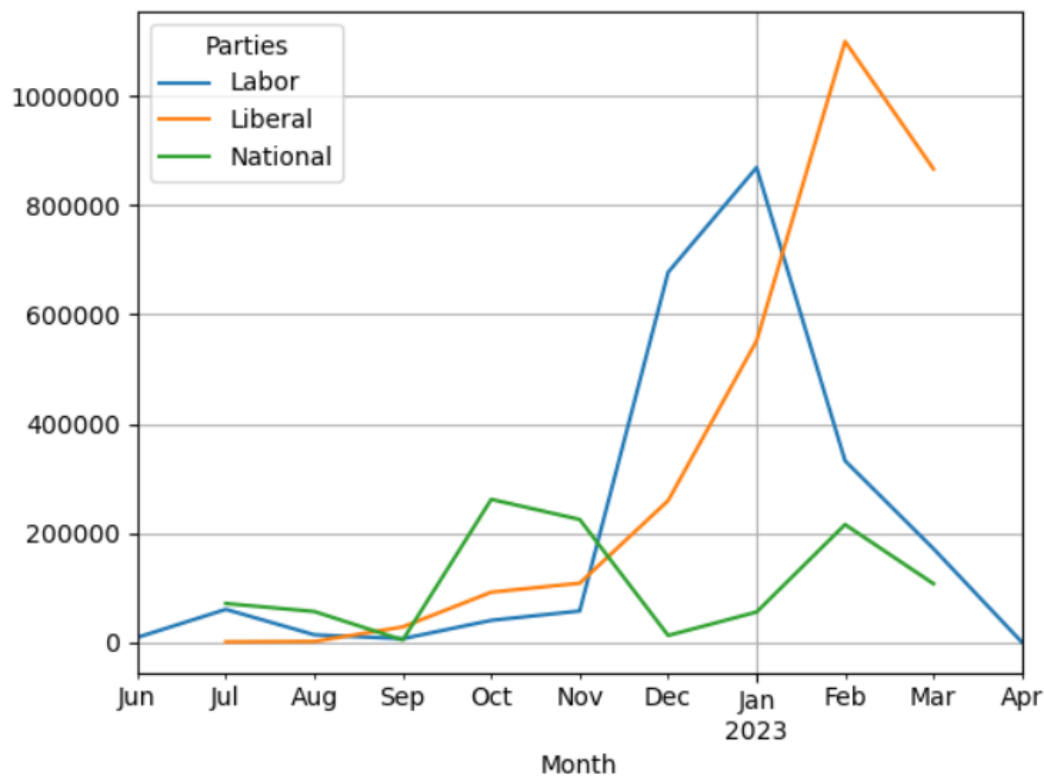


Figure 4 : Monthly expenditure trends by party

Spending increases as the election period approaches within the analysed period (July 2022 to March 2023). The Liberal Party shows a sharp peak of AUD 1.1M in February, while Labor exhibits earlier peaks. These patterns suggest different campaign timing strategies across parties.

9.5 Key Insights and Limitations

The EDA highlights several findings:

- Electoral spending patterns differ significantly across parties.
- Expenditure data is highly skewed due to large transactions.
- Category-level insights may be influenced by large transactions.
- Mean-based comparisons may not accurately represent typical spending behaviour.

Additionally, the analysis focuses on three major parties (Liberal, Labor, and National) and is limited to the period from July 2022 to March 2023, which may not capture the full election cycle. These findings inform the approach of this project, particularly the need to account for outliers and improve interpretability through structured analysis.

References:

- [1] "New data reveals record spending ahead of 2025 election – but taxpayer funding will increase for 2028," The Australia Institute, Feb. 2, 2026. [Online]. Available: <https://australiainstitute.org.au/post/new-data-reveals-record-spending-ahead-of-2025-election-but-taxpayer-funding-will-increase-for-2028/>
- [2] "What is electoral expenditure?" NSW Electoral Commission, 2021. [Online]. Available: <https://elections.nsw.gov.au/electoral-funding/electoral-expenditure/what-is-electoral-expenditure>
- [3] M. Branson and W. George, "Misleading Graphs." [Online]. Available: <https://web.stevenson.edu/mbranson/m4tp/version1/fake-news-misleading-graphs.html>
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- [5] D. Coates, "The effects of campaign spending on electoral outcomes: A data envelopment analysis," *Public Choice*, vol. 99, pp. 15–37, 1999.
- [6] B. Reid, "Electoral studies and data-driven analysis: A Victorian case study," *Australasian Parliamentary Review*, vol. 39, no. 1, pp. 88–111, 2024. [Online]. Available: https://www.researchgate.net/publication/380720317_Electoral_Studies_and_data-driven_analysis_A_Victorian_case_study

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